



ST BENEDICT'S

SUBJECT	Mathematics	PAPER	Functions and Statistics Section B
GRADE	11	DATE	7 June 2023
EXAMINER NAME	M Brown	MARKS	35
TEACHER	Memo	MODERATOR	Mrs Povall, Mr Hayward, Mrs Meredith
		DURATION	40 minutes

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Scatter Plots	9	
2	Functions	7	
3	Exponential Functions	10	
4	Hyperbolic Functions	9	
TOTAL		35	

INSTRUCTIONS:

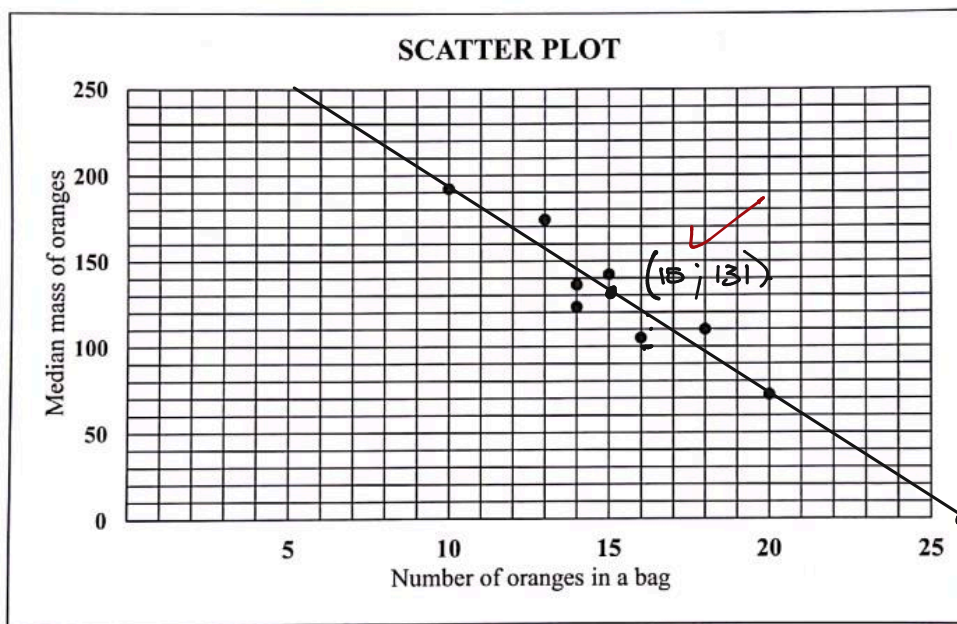
1. This paper consists of 4 questions and 7 pages. A page for extra working has been provided, if this page is used, label the question accurately. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer all questions on the question paper and hand it in at the end of the test.
4. It is in your interest to write legibly and present your work neatly.
5. Clearly show **ALL** calculations, diagrams, graphs, etc. that you have used in determining your answers. **Answers only will NOT necessarily be awarded full marks.**
6. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
7. Ensure that your calculator is in **DEGREE** mode.
8. Round off your answers to **ONE** decimal place unless otherwise stated.

QUESTION 1

9 MARKS

A student is investigating the number of oranges in a bag in relation to the median mass of the oranges in the same bag. The findings are recorded in the table below.

Number of oranges in a bag	18	16	20	15	14	13	14	10
Median mass of oranges in the same bag (to the nearest gram)	110	105	72	142	123	174	136	192



- a) Determine the equation of the least squares regression line for the data to 2 decimal digits. (2)

$$y = 307,20 - 11,7x$$

- b) Write down the correlation coefficient of the data, and comment on the strength of the relationship between the number of oranges in the bag and the median mass of the oranges. (3)

$$r = -0,933$$

∴ strong, negative, linear relationship

- c) Draw the least squares regression line on the scatter plot. $(\bar{x}; \bar{y})$ (2)

- d) Determine the possible median mass of oranges in a bag, if there are 12 oranges in that bag. (2)

$$y = 307,20 - 11,7(12) = 166,8 \text{ grams}$$

QUESTION 2

7 MARKS

- a) Determine the equation of the straight line passing through $(-2; 5)$ and parallel to the line $x + 2y - 6 = 0$. (3)

$$2y = -x + 6$$

$$y = -\frac{x}{2} + 3$$

$$\therefore y = -\frac{x}{2} + c$$

$$5 = -\frac{(-2)}{2} + c$$

$$\therefore c = 4$$

$$\therefore y = -\frac{x}{2} + 4$$

- b) Give the equation of a quadratic function in the form $y = ax^2 + bx + c$ if it is given that: (4)

- the range of f is: $y \geq -4$

- the domain is: $x \in \mathbb{R}$

- Zero points are $(3; 0)$, $(-1; 0)$ and $(0; -3)$.

$$y = a(x+1)(x-3)$$

$$-3 = a(-3)$$

$$a = 1$$

$$\therefore y = x^2 - 2x - 3$$

$$AS: x = 1$$

$$T.P (1; 4)$$

$$\therefore y = a(x-1)^2 - 4$$

$$-3 = a(-1)^2 - 4$$

$$1 = a$$

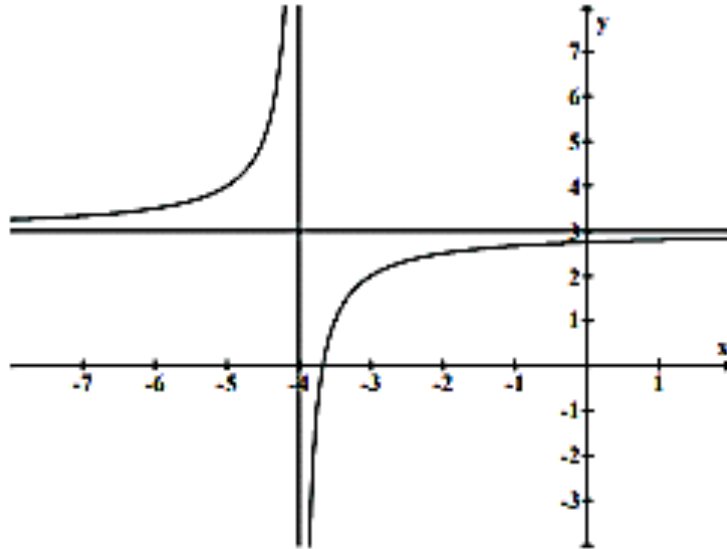
$$\therefore y = (x^2 - 2x + 1) - 4$$

$$= x^2 - 2x - 3$$

QUESTION 3

10 MARKS

Sketched below is the graph of $f(x) = \frac{a}{x+p} + q$.



- a) Write down the values of p and q . (2)

$$p = 4 \quad \checkmark \quad A$$

$$q = 3 \quad \checkmark \quad A$$

- b) Calculate the value of a if $T(-2; 2,5)$ lies on the graph of f . (2)

$$2,5 = \frac{a}{-2+4} + 3 \quad \checkmark \quad m$$

$$-0,5 \times 2 = a$$

$$a = -1 \quad \checkmark \quad A$$

- c) Write down the domain of f . (1)

$$x \in \mathbb{R}; x \neq -4 \quad \checkmark \quad A$$

- d) Write down the equation for the lines of symmetry of f . (2)

$$y = \frac{-1}{x+4} + 3$$

$$\therefore y = x+7 \quad \checkmark \quad A \quad \text{and} \quad y = -x-1 \quad \checkmark \quad A$$

- e) Write down the range of $f(x) - 1$.

(2)

$y \in \mathbb{R}, y \neq 2$. $\checkmark$ A ($y \neq 2$) $\checkmark$ A ($y \in \mathbb{R}$).

- f) Describe the transformation of f to g if $g(x) = -f(x)$.

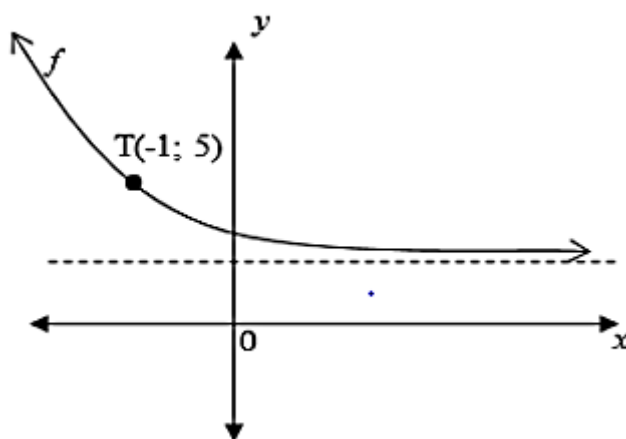
(1)

Reflection over x -axis $\checkmark$

QUESTION 4

9 MARKS

On the sketch below, $f(x) = b^x + 2$ is drawn, and $T(-1; 5)$ is a point on f .



- a) Calculate the value of b .

(3)

$5 = b^{-1} + 2$ $\checkmark$ A.
 $3 = b^{-1}$
 $\therefore b = \frac{1}{3}$ $\checkmark$ m A.

- b) Calculate the coordinates of the y -intercept of f .

(2)

$y = 1 + 2 = 3$. $(0; 3)$ $\checkmark$ A. $\checkmark$ A.

- c) Determine the equation of h , the reflection of f about the y -axis. (2)

$$h(x) = 3^x + 2.$$

- d) The point $(x; 11)$ lies on f . Calculate the value of x . (2)

$$11 = \left(\frac{1}{3}\right)^x + 2.$$

$$9 = \left(\frac{1}{3}\right)^x.$$

$$3^2 = 3^{-x}$$

$$x = -2.$$

EXTRA WORKING: